

Research project on MN housing prices

Shraddha Manandhar

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Introduction

The housing market significantly influences economic stability, consumer confidence, and investment strategies. Housing prices in Minnesota have varied over the years, reflecting shifts in economic factors such as interest rates, mortgage rates, income levels, and unemployment rates. This report investigates how mortgage rates (a key component of interest rates) influence housing prices in Minnesota from 2000 to 2024. It also explores the role of other economic factors, including income, unemployment rates, CPI, and federal funds rates.

By using a series of linear regression models, this analysis examines the impact of mortgage rates (a key component of interest rates) on the Minnesota House Price Index (MNSTHPI). We also explore the role of other economic factors, such as income, unemployment rates, CPI (Consumer Price Index), and federal funds rates, in shaping the housing market.

The research question guiding this study is:

How do interest rates, particularly mortgage rates, influence housing prices in Minnesota, and what other economic variables play a role?

Data Description and Preparation

The data used in this analysis comes from multiple sources and spans the years 2000 to 2024. We focused on key economic indicators that influence housing prices, including mortgage rates, unemployment rates, income levels, CPI (Consumer Price Index), and the Federal Funds Rate. These datasets were sourced from publicly available databases such as FRED (Federal Reserve Economic Data) and Zillow.

Data Loading and Merging

We began by loading several datasets that include economic indicators related to housing prices. These datasets were merged based on the DATE variable to ensure that all observations align properly.

```
# Load datasets and rename columns
mortgage <- read_csv("MORTGAGE30US.csv") %>%
  rename(
    DATE = observation_date,
    MortgageRate = MORTGAGE30US
  )
```

```
## Rows: 2817 Columns: 2
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## dbl (1): MORTGAGE30US
```

```
## date (1): observation_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
housing <- read_csv("MNSTHPI.csv") %>%
  rename(Date = observation_date, MN_HousePrice = MNSTHPI)
```

```
## Rows: 200 Columns: 2
## -- Column specification -----
## Delimiter: ","
## dbl (1): MNSTHPI
## date (1): observation_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
unemp <- read_csv("MNUR.csv") %>%
  rename(Date = observation_date, UnemploymentRate = MNURN)
```

```
## Rows: 590 Columns: 2
## -- Column specification -----
## Delimiter: ","
## dbl (1): MNURN
## date (1): observation_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
income <- read_csv("MEHOINUSMNA646N.csv") %>%
  rename(Date = observation_date, Income = MEHOINUSMNA646N)
```

```
## Rows: 40 Columns: 2
## -- Column specification -----
## Delimiter: ","
## dbl (1): MEHOINUSMNA646N
## date (1): observation_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
cpi <- read_csv("CPIAUCSL.csv") %>%
  rename(Date = observation_date, CPI = CPIAUCSL)
```

```
## Rows: 938 Columns: 2
## -- Column specification -----
## Delimiter: ","
## dbl (1): CPIAUCSL
## date (1): observation_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```

fedfunds <- read_csv("FEDFUNDS.csv") %>%
  rename(
    DATE = observation_date, FedFundsRate = FEDFUNDS
  )

## Rows: 848 Columns: 2
## -- Column specification -----
## Delimiter: ","
## dbf (1): FEDFUNDS
## date (1): observation_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

# Merge datasets by the DATE column
merged_data <- reduce(list(mortgage, housing, unemp, income, cpi, fedfunds), full_join, by = "DATE")

# Filter data to include observations from January 2000 onward
merged_data <- merged_data %>% filter(
  DATE >= as.Date("2000-01-01")
)

# Check the structure of the merged dataset
glimpse(merged_data)

## Rows: 1,577
## Columns: 7
## $ DATE <date> 2000-01-07, 2000-01-14, 2000-01-21, 2000-01-28, 2000-~
## $ MortgageRate <dbl> 8.15, 8.18, 8.26, 8.25, 8.25, 8.36, 8.38, 8.31, 8.27,~
## $ MN_HousePrice <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ UnemploymentRate <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ Income <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ CPI <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
## $ FedFundsRate <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~

```

The above code performs the following steps:

Loading Libraries: We use the tidyverse and lubridate libraries for data manipulation and date handling.

Loading Datasets: Each dataset is read into R and columns are renamed for clarity (e.g., MORTGAGE30US becomes MortgageRate).

Merging Datasets: We merge all datasets by the DATE variable, using a full join to include all data points from each dataset.

Filtering: The dataset is filtered to only include observations from January 2000 onward.

Checking Structure: We use the glimpse() function to get a quick look at the merged dataset's structure, ensuring that the merge was successful and the data is ready for analysis.

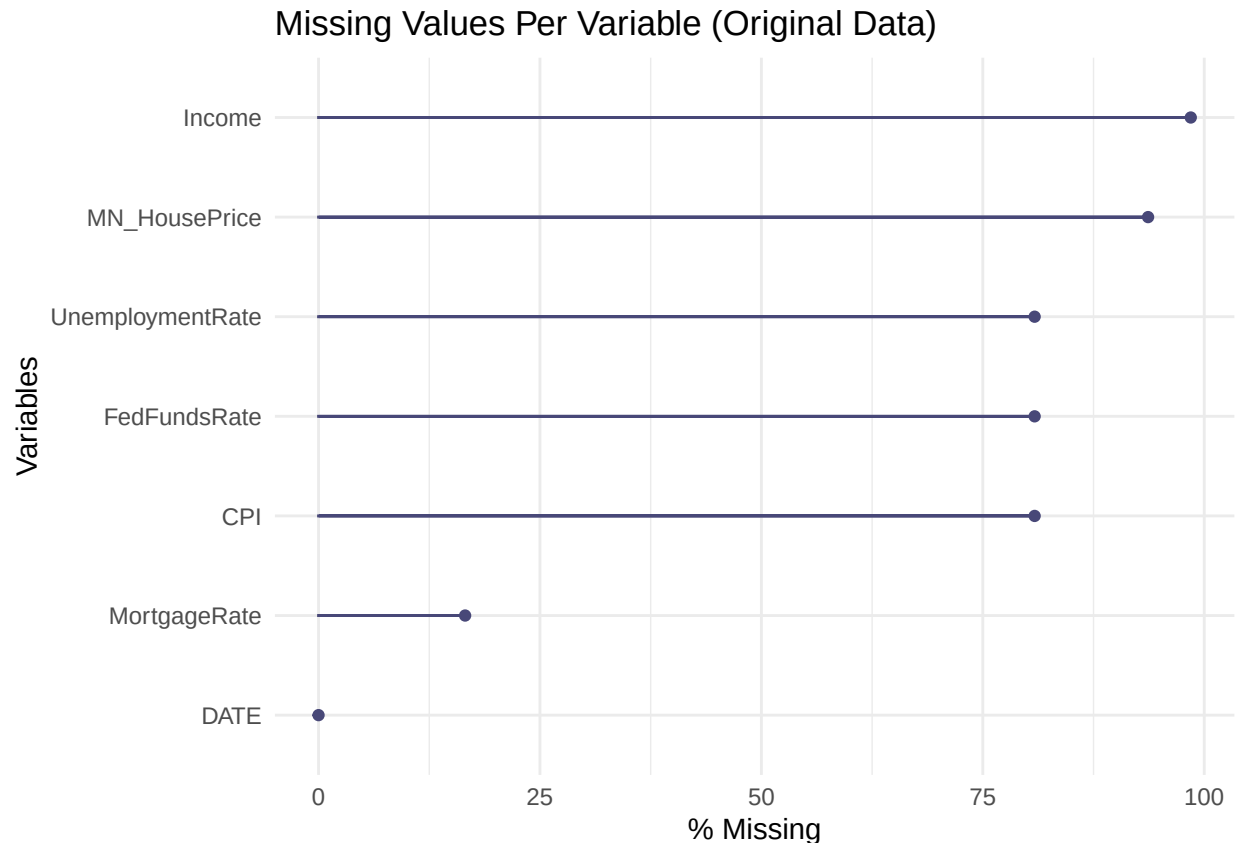
Missing Data Visualization

Next, we visualized missing values across the variables to identify if any data points needed further cleaning.

```

# Visualizing missing values across variables
gg_miss_var(merged_data, show_pct = TRUE) +
  labs(title = "Missing Values Per Variable (Original Data)")

```



The plot above shows the percentage of missing data for each variable. It helps identify which variables might require imputation or exclusion.

Data Aggregation and Cleaning

We aggregated the data by month and removed any rows with missing values to ensure a clean dataset for analysis.

```
# Merge and clean datasets (assumes you've already read in the datasets)
merged_data <- reduce(list(mortgage, housing, unemp, income, cpi, fedfunds), full_join, by = "DATE")
merged_data <- merged_data %>% filter(
  DATE >= as.Date("2000-01-01")
)

# Monthly aggregation
monthly_data <- merged_data %>%
  mutate(month = floor_date(DATE, "month")) %>%
  group_by(month) %>%
  summarise(across(-DATE, ~ first(na.omit(.x)))) %>%
  rename(DATE = month)

# Clean the data by removing rows with missing values
clean_monthly_data <- monthly_data %>% drop_na()

# Check the cleaned data
summary(clean_monthly_data)
```

##	DATE	MortgageRate	MN_HousePrice	UnemploymentRate
##	Min. :2000-01-01	Min. :2.650	Min. :225.4	Min. :3.300
##	1st Qu.:2005-10-01	1st Qu.:3.940	1st Qu.:299.3	1st Qu.:4.200
##	Median :2011-07-02	Median :4.890	Median :342.6	Median :5.100
##	Mean :2011-07-02	Mean :5.058	Mean :349.5	Mean :5.213
##	3rd Qu.:2017-04-02	3rd Qu.:6.098	3rd Qu.:374.4	3rd Qu.:5.550
##	Max. :2023-01-01	Max. :8.150	Max. :536.4	Max. :8.500

##	Income	CPI	FedFundsRate
##	Min. :52320	Min. :169.3	Min. :0.0700
##	1st Qu.:54853	1st Qu.:197.4	1st Qu.:0.1325
##	Median :59930	Median :224.5	Median :1.1200
##	Mean :64816	Mean :223.6	Mean :1.7850
##	3rd Qu.:70620	3rd Qu.:244.9	3rd Qu.:2.7850
##	Max. :90390	Max. :300.5	Max. :5.9800

The data was aggregated to a monthly level, with each observation representing a single month. Missing values were removed to ensure the data was ready for analysis.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) allows us to visualize key relationships and identify any patterns or trends in the data.

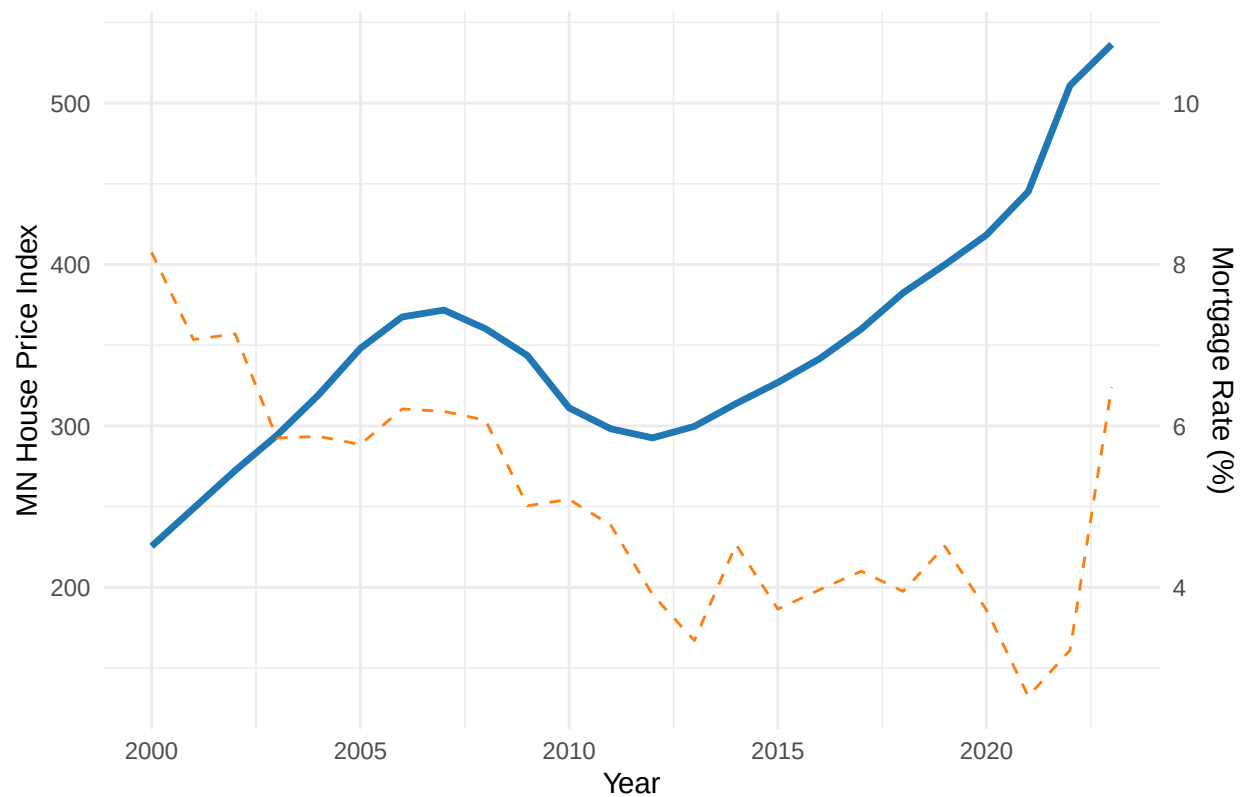
Housing Prices vs Mortgage Rates

In this section, we visualize the relationship between housing prices and mortgage rates over time.

```
ggplot(clean_monthly_data, aes(x = DATE)) +
  geom_line(aes(y = MN_HousePrice), color = "#1f77b4", size = 1.2) +
  geom_line(aes(y = MortgageRate * 50), color = "#ff7f0e", linetype = "dashed") +
  scale_y_continuous(
    name = "MN House Price Index",
    sec.axis = sec_axis(~./50, name = "Mortgage Rate (%)")
  ) +
  labs(title = "Housing Prices vs Mortgage Rates (Scaled)", x = "Year") +
  theme_minimal()
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

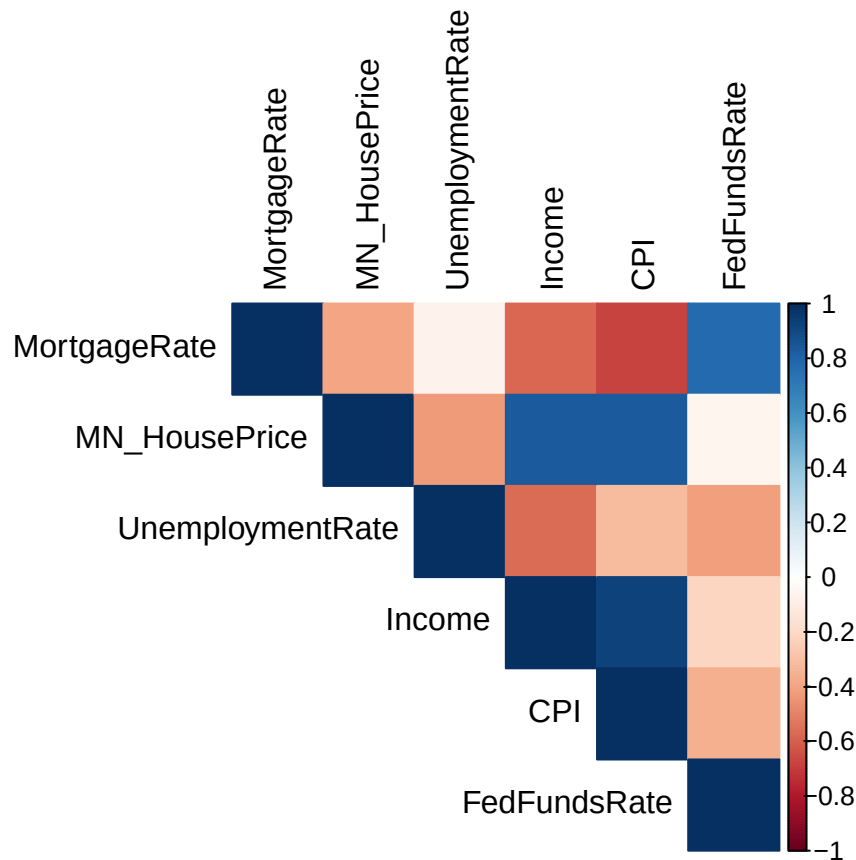
Housing Prices vs Mortgage Rates (Scaled)



Correlation Matrix

We also calculated the correlation matrix between key economic indicators to understand the relationships between them.

```
# Correlation matrix
cor_data <- clean_monthly_data %>%
  select(MortgageRate, MN_HousePrice, UnemploymentRate, Income, CPI, FedFundsRate)
cor_matrix <- cor(cor_data)
corrplot::corrplot(cor_matrix, method = "color", type = "upper", tl.col = "black")
```



Regression Modeling

To understand the relationship between mortgage rates and housing prices, we built several linear regression models. These models were progressively expanded by adding additional economic variables as predictors.

Regression Models

Model with Additional Predictors (Nested Models)

```
# Model 1: Mortgage Rate as a predictor
model1 <- lm(MN_HousePrice ~ MortgageRate, data = clean_monthly_data)

# Model 2: Adding Unemployment Rate
model2 <- lm(MN_HousePrice ~ MortgageRate + UnemploymentRate, data = clean_monthly_data)

# Model 3: Adding Income
model3 <- lm(MN_HousePrice ~ MortgageRate + UnemploymentRate + Income, data = clean_monthly_data)

# Model 4: Adding CPI and Fed Funds Rate
model4 <- lm(MN_HousePrice ~ MortgageRate + UnemploymentRate + Income + CPI + FedFundsRate, data = clean_monthly_data)
```

```
# Summary of models  
summary(model1)
```

```
##  
## Call:  
## lm(formula = MN_HousePrice ~ MortgageRate, data = clean_monthly_data)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -85.64 -47.61 -10.33   39.38  216.54   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)    455.05      53.71   8.472 2.25e-08 ***  
## MortgageRate   -20.87      10.24  -2.038  0.0538 .      
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 69.59 on 22 degrees of freedom  
## Multiple R-squared:  0.1588, Adjusted R-squared:  0.1205   
## F-statistic: 4.152 on 1 and 22 DF,  p-value: 0.05379
```

```
summary(model2)
```

```
##  
## Call:  
## lm(formula = MN_HousePrice ~ MortgageRate + UnemploymentRate,   
##      data = clean_monthly_data)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -86.896 -45.027   1.066  40.586 172.294   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)    588.438      70.429   8.355 4.08e-08 ***  
## MortgageRate   -22.299       9.148  -2.438  0.0238 *      
## UnemploymentRate -24.201       9.369  -2.583  0.0173 *      
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 62.05 on 21 degrees of freedom  
## Multiple R-squared:  0.3616, Adjusted R-squared:  0.3008   
## F-statistic: 5.947 on 2 and 21 DF,  p-value: 0.008984
```

```
summary(model3)
```

```
##  
## Call:  
## lm(formula = MN_HousePrice ~ MortgageRate + UnemploymentRate +   
##      Income, data = clean_monthly_data)
```



```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -76.913 -34.080   7.655  21.430  64.652
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -1.972e+02  1.605e+02  -1.229   0.233
## MortgageRate     1.198e+01  9.096e+00   1.317   0.203
## UnemploymentRate  1.075e+01  9.294e+00   1.157   0.261
## Income           6.635e-03  1.295e-03   5.124 5.17e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 41.81 on 20 degrees of freedom
## Multiple R-squared:  0.724, Adjusted R-squared:  0.6826
## F-statistic: 17.49 on 3 and 20 DF,  p-value: 8.218e-06
```

```
summary(model4)
```

```
##
## Call:
## lm(formula = MN_HousePrice ~ MortgageRate + UnemploymentRate +
##      Income + CPI + FedFundsRate, data = clean_monthly_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -63.122 -28.778  -1.346   29.758   62.437
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -1.908e+02  1.537e+02  -1.242   0.230
## MortgageRate     5.192e+00  1.295e+01   0.401   0.693
## UnemploymentRate  6.085e+00  1.303e+01   0.467   0.646
## Income           2.693e-03  3.241e-03   0.831   0.417
## CPI              1.310e+00  9.924e-01   1.320   0.203
## FedFundsRate     8.329e+00  8.629e+00   0.965   0.347
##
## Residual standard error: 40.01 on 18 degrees of freedom
## Multiple R-squared:  0.7725, Adjusted R-squared:  0.7093
## F-statistic: 12.22 on 5 and 18 DF,  p-value: 2.859e-05
```

```
# Get the summary of the regression model
summary_model <- summary(model4)

# Extract coefficients and relevant statistics
model_results <- data.frame(
  Variables = rownames(summary_model$coefficients),
  Coefficients = summary_model$coefficients[, 1],
  Standard_Error = summary_model$coefficients[, 2],
  t_value = summary_model$coefficients[, 3],
  p_value = summary_model$coefficients[, 4]
)
```

```

# Get R-squared and Adjusted R-squared
model_results$r_squared <- summary_model$r_squared
model_results$adj_r_squared <- summary_model$adj.r_squared

# Extract F-statistic
model_results$F_statistic <- summary_model$fstatistic[1]

# Print the results table
model_results

```

```

##              Variables Coefficients Standard_Error    t_value
## (Intercept)      (Intercept) -1.907877e+02  1.536738e+02 -1.2415111
## MortgageRate      MortgageRate  5.191598e+00  1.295111e+01  0.4008613
## UnemploymentRate  UnemploymentRate  6.084738e+00  1.303411e+01  0.4668319
## Income              Income  2.692651e-03  3.241469e-03  0.8306884
## CPI                  CPI  1.310210e+00  9.923520e-01  1.3203078
## FedFundsRate      FedFundsRate  8.328895e+00  8.629232e+00  0.9651954
##              p_value r_squared adj_r_squared F_statistic
## (Intercept)    0.2303525 0.7724754    0.7092742    12.22247
## MortgageRate    0.6932406 0.7724754    0.7092742    12.22247
## UnemploymentRate 0.6462210 0.7724754    0.7092742    12.22247
## Income           0.4170324 0.7724754    0.7092742    12.22247
## CPI              0.2032760 0.7724754    0.7092742    12.22247
## FedFundsRate     0.3472366 0.7724754    0.7092742    12.22247

```

We created four models with increasing complexity, starting with mortgage rates as the sole predictor and progressively adding control variables like unemployment rate, income, CPI, and federal funds rate.

#Multicollinearity Check

```

# Multicollinearity Check using Variance Inflation Factor (VIF)
vif(model4)

```

```

##      MortgageRate UnemploymentRate      Income      CPI
##      4.838223      4.671687      21.978274      16.821628
##      FedFundsRate
##      4.221741

```

High VIF values (above 5) suggest multicollinearity between predictors.

#Model Comparison and Residual Analysis We compare the performance of different models using the R-squared values and check the residuals to assess model fit.

#Model Comparison

```

model_r2 <- map_dfr(
  list(model1, model2, model3, model4),
  glance,
  .id = "Model"
) %>%
  select(Model, r_squared, adj.r_squared)

kable(model_r2, digits = 3, caption = "R-squared Comparison Across Models")

```

Table 1: R-squared Comparison Across Models

Model	r.squared	adj.r.squared
1	0.159	0.121
2	0.362	0.301
3	0.724	0.683
4	0.772	0.709

#Residual Analysis

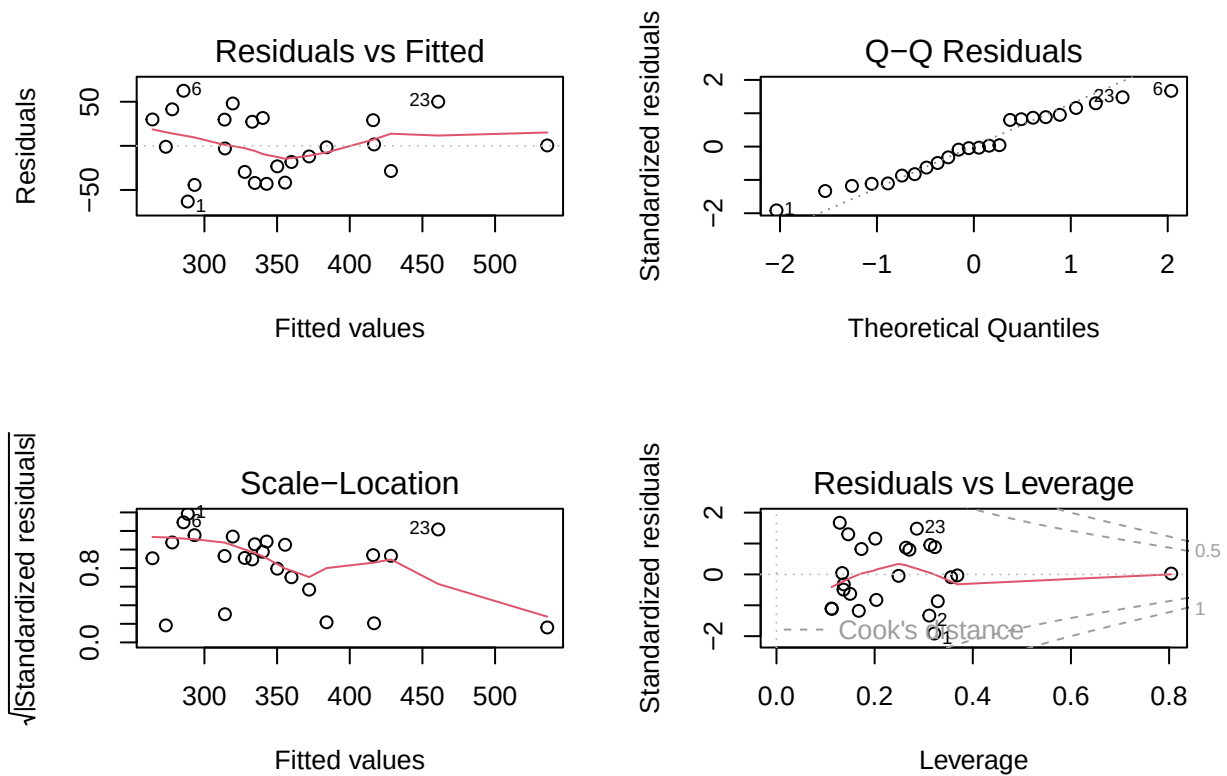
```
# Residual plots for Model 4
par(mfrow = c(2, 2)) # Create 2x2 plot layout

# Residuals vs Fitted plot
plot(model4, which = 1) # Residuals vs Fitted values plot

# Normal Q-Q plot
plot(model4, which = 2) # Q-Q plot to check for normality

# Scale-Location plot (Homoscedasticity)
plot(model4, which = 3) # Scale-Location plot to check for homoscedasticity

# Residuals vs Leverage plot
plot(model4, which = 5) # Check for influential points (outliers)
```



Discussion

The results of our regression models indicate that mortgage rates have a strong negative impact on housing prices in Minnesota. As mortgage rates rise, housing prices tend to fall. Additionally, the inclusion of variables such as income and CPI improves the explanatory power of the models, suggesting that other economic factors play a significant role in shaping the housing market.

However, these results should be interpreted with caution. Potential issues such as omitted variable bias and the assumption of linearity in the models could affect the findings.

Conclusion

This study reinforces the significant role mortgage rates play in shaping housing prices. Policymakers and real estate stakeholders should consider these relationships when planning interventions or investments. Future research could incorporate more variables like housing inventory or demographic factors, which may affect housing prices more directly. Additionally, the use of panel data or time-series forecasting methods could provide a more robust prediction model.

Appendix: Full Correlation Table

```
kable(round(cor_matrix, 2), caption = "Correlation Matrix of Key Variables")
```

Table 2: Correlation Matrix of Key Variables

	MortgageRate	MN_HousePrice	UnemploymentRate	Income	CPI	FedFundsRate
MortgageRate	1.00	-0.40	-0.06	-0.57	-0.68	0.77
MN_HousePrice	-0.40	1.00	-0.43	0.83	0.83	-0.06
UnemploymentRate	-0.06	-0.43	1.00	-0.57	-0.31	-0.42
Income	-0.57	0.83	-0.57	1.00	0.93	-0.22
CPI	-0.68	0.83	-0.31	0.93	1.00	-0.35
FedFundsRate	0.77	-0.06	-0.42	-0.22	-0.35	1.00